TABLEAU PORTFOLIO PROJECT

GROUP D: SPC SALES PER CAPITA -- AUSTRALIAN RETAIL ANALYSIS

SPC Sales Per Capita Analysis

Multi-region time series, forecasting with confidence bands, and tabbed story presentation

Tableau | Time Series | Forecasting | Box Plot | Trend Lines | Story | Faceted Layout | Australian Reta

Project Summary

A 5-view Tableau workbook analyzing Australian retail Sales Per Capita (SPC) across 4 regions (NSW, QLD, VIC, WA) from 2000 to 2019. Includes competitor box-and-whisker analysis, industry turnover time series, per-capita value trends, forecast projections with confidence bands extending to 2019, and a fully navigable tabbed Tableau Story presentation. Demonstrates advanced forecasting, multi-panel faceted layout, and Tableau Story authoring.

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1 PROJECT OVERVIEW

Workbook:	SPC Sales Per Capita -- Australian Retail
Views:	5 (Competitors Analysis, Industry Analysis, SPC Value, SPC Forecast, SPC Presentation Story)
Data Source:	Australian retail dataset -- turnover $M and SPC $ by region, 2000-2019
Regions Covered:	NSW (New South Wales), QLD (Queensland), VIC (Victoria), WA (Western Australia)
Historical Range:	2000 to 2015 (actuals)
Forecast Range:	2016 to 2019 (Tableau built-in forecasting engine)
Chart Types:	Box-and-whisker + dot overlay, Faceted time series, Forecast with confidence bands, Tableau Story
Business Use Case:	Regional retail performance benchmarking and forward-looking sales planning

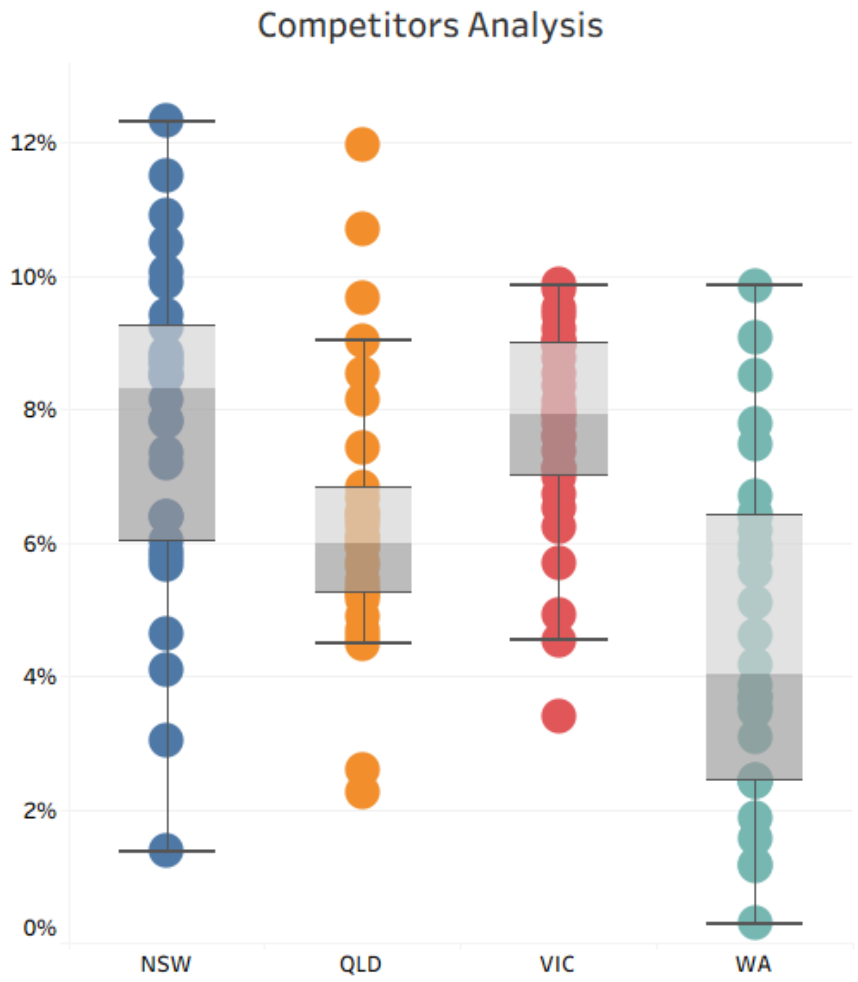
2 BUSINESS CONTEXT

Retail analysts and executives need to answer two questions simultaneously: how are our regions performing relative to competitors (competitive benchmarking), and where are our sales per capita trends heading (forward planning)? This workbook answers both through a structured 4-view analysis capped by a Tableau Story that presents the full narrative in a tab-navigable presentation format suitable for C-suite or board-level delivery.

- NSW consistently leads in both Turnover $M (up to ~600M peak) and SPC $ (up to ~100), with strong upward trend across the full 15-year history.
- VIC shows the steepest growth trajectory of all regions in SPC $, starting lowest and approaching NSW by 2015.
- QLD turnover is second highest but SPC $ is essentially flat (~40-50), suggesting population growth is outpacing revenue growth.
- WA shows strong absolute SPC performance with minimal trend growth -- a mature, stable market.
- Competitor Analysis box plots reveal NSW has the widest variance in competitor market share (1-13%), while WA has the tightest distribution, implying fewer competitive disruptions.

3 VIEW 1 -- COMPETITORS ANALYSIS: BOX-AND-WHISKER BY REGION

Box-and-whisker chart with individual data point dot overlay, showing distribution of competitor market share percentages across the 4 Australian retail regions. Each box shows Q1-Q3 range with median line; whiskers extend to min/max; individual dots show every competitor observation.

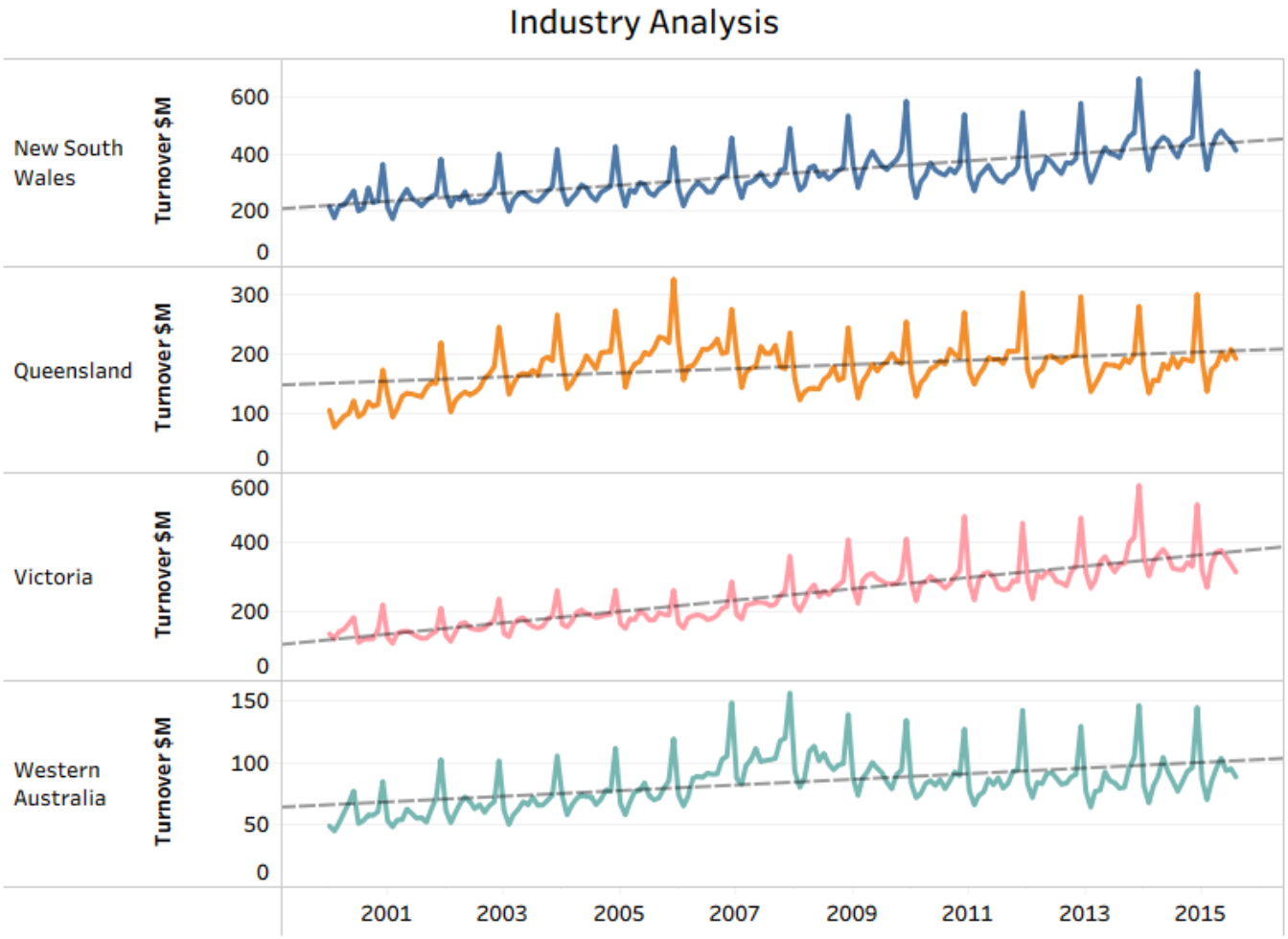


Competitors Analysis -- Box-and-whisker + dot overlay by region (NSW, QLD, VIC, WA). Y-axis: market share %. NSW median ~8%, QLD ~6%, VIC ~8%, WA ~4%.

Tableau technique: Box-and-whisker plots created via Analytics pane > Box Plot added to an existing dot-chart. Circle marks overlay the box to show individual competitor data points. Color encodes region.

4 VIEW 2 -- INDUSTRY ANALYSIS: TURNOVER $M BY REGION (2000-2015)

Faceted time series showing monthly Turnover $M for each of 4 regions on independent Y-axes. Dashed grey trend lines show the overall growth direction. Strong seasonal peaks are visible in all regions (December spikes in NSW up to ~600M). The faceted layout enables direct region-by-region comparison without scale distortion.

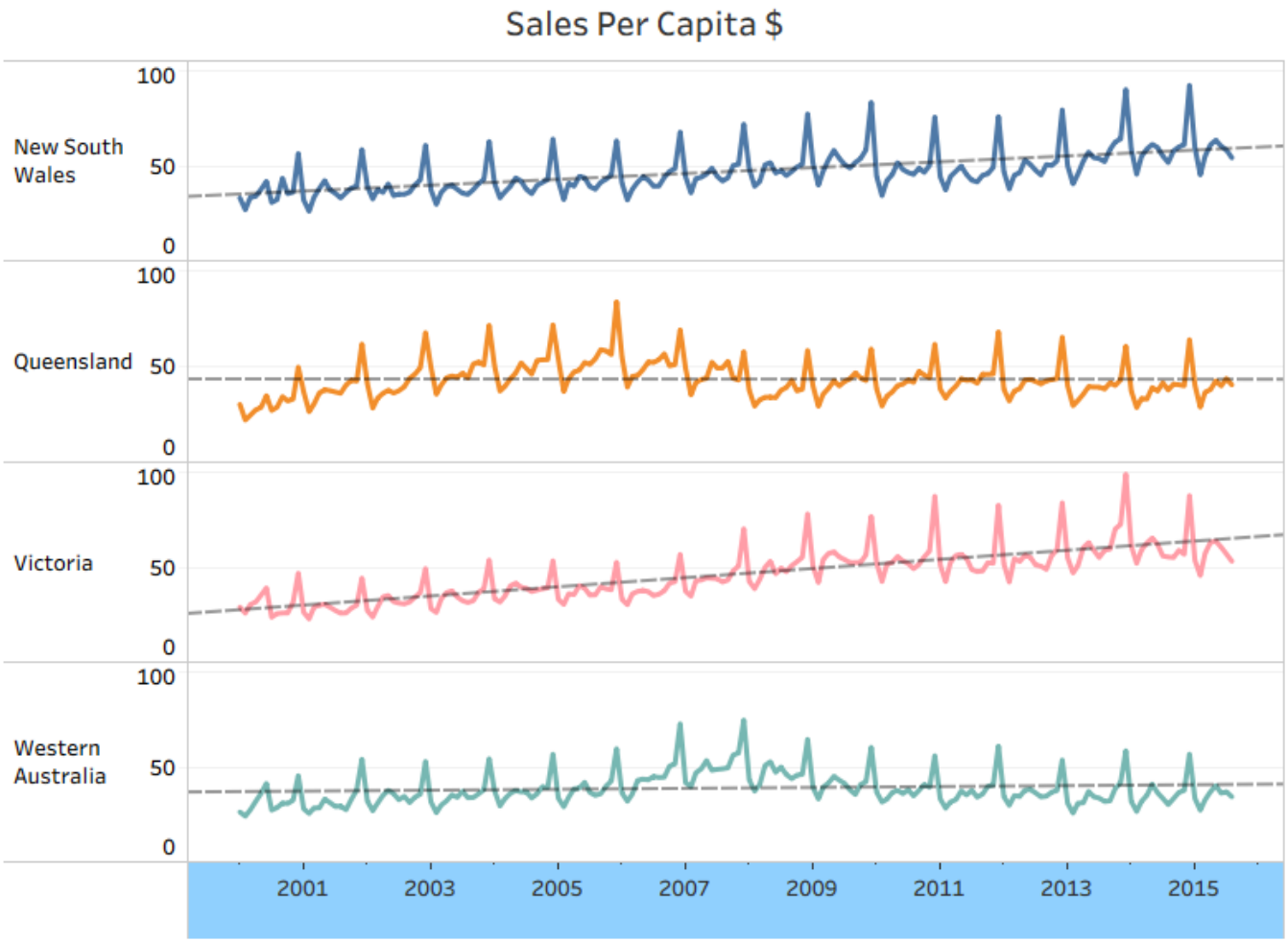


Industry Analysis -- Faceted time series: Turnover $M by region (NSW, QLD, VIC, WA) from 2000-2015. Dashed trend lines show growth trajectory. NSW peaks ~600M, WA ~130M.

Tableau technique: Row shelf contains [Region] as a discrete dimension, creating 4 separate horizontal panels. Each panel shares the same X-axis (date) but has an independent Y-axis scale. Trend lines added via Analytics pane > Trend Line > Linear.

5 VIEW 3 -- SALES PER CAPITA $: PER-CAPITA VALUE TRENDS (2000-2015)

Same faceted layout as Industry Analysis but normalized to Sales Per Capita $ -- eliminating population size as a confounding factor. This reveals VIC's steeper per-capita growth vs. its lower absolute turnover, and QLD's flat per-capita trend despite high absolute revenue.

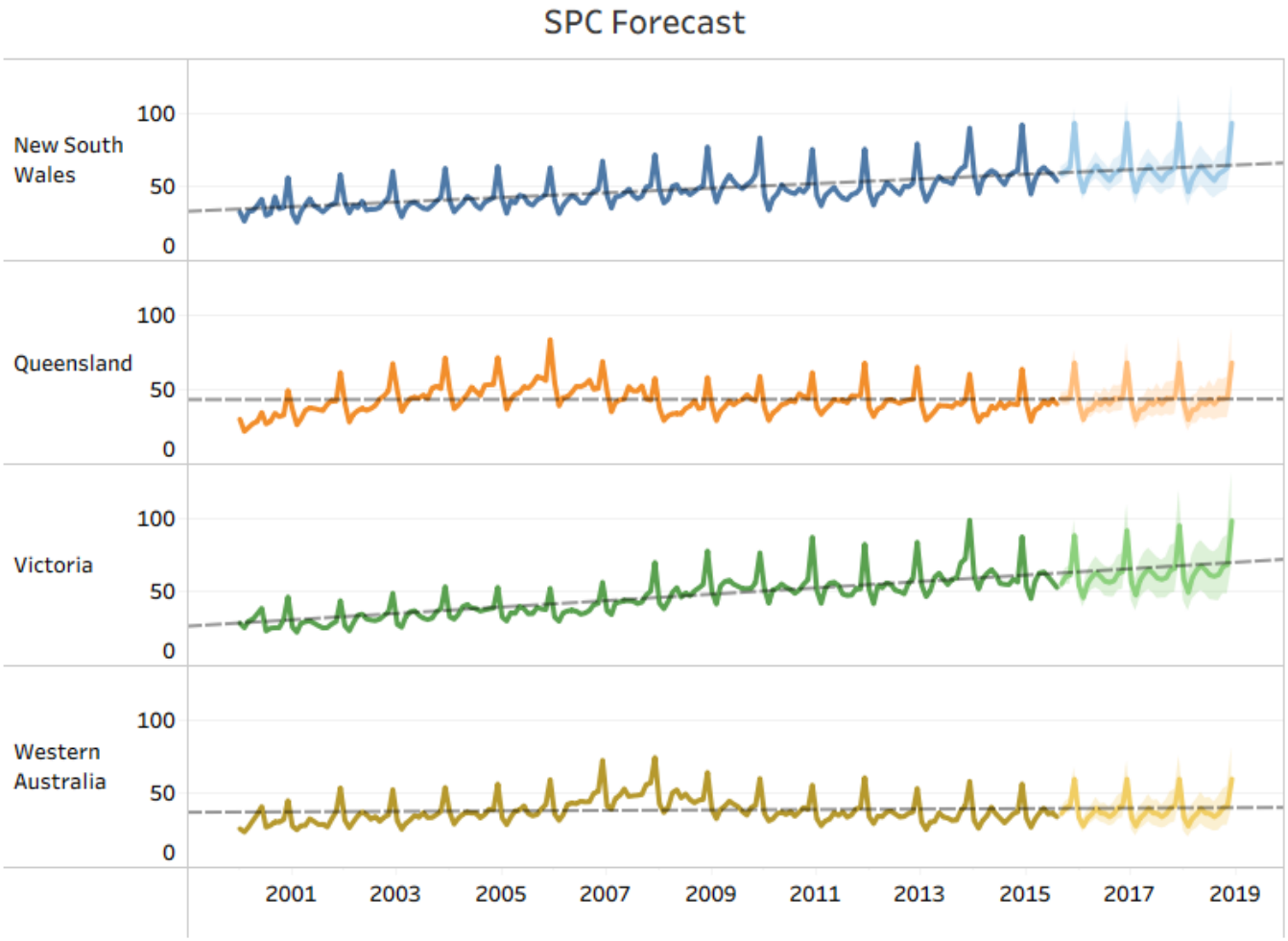


Sales Per Capita $ -- Faceted time series: SPC $ by region 2000-2015. NSW leads (~60-100 $), VIC steepest growth, QLD flat at ~40-50 $, WA stable ~40 $.

> *Key analytical insight: QLD has the second highest absolute Turnover but nearly the flattest SPC growth -- indicating population growth is absorbing revenue gains. VIC has lower absolute turnover but the steepest per-capita growth -- a more efficient market expansion story.*

6 VIEW 4 -- SPC FORECAST: CONFIDENCE BAND PROJECTIONS TO 2019

The historical SPC time series is extended to 2019 using Tableau's built-in exponential smoothing forecast engine. Shaded confidence bands around each region's forecast show the range of plausible outcomes. The forecast period (2016-2019) is visually distinct from the historical period (2000-2015) by the shaded envelope.



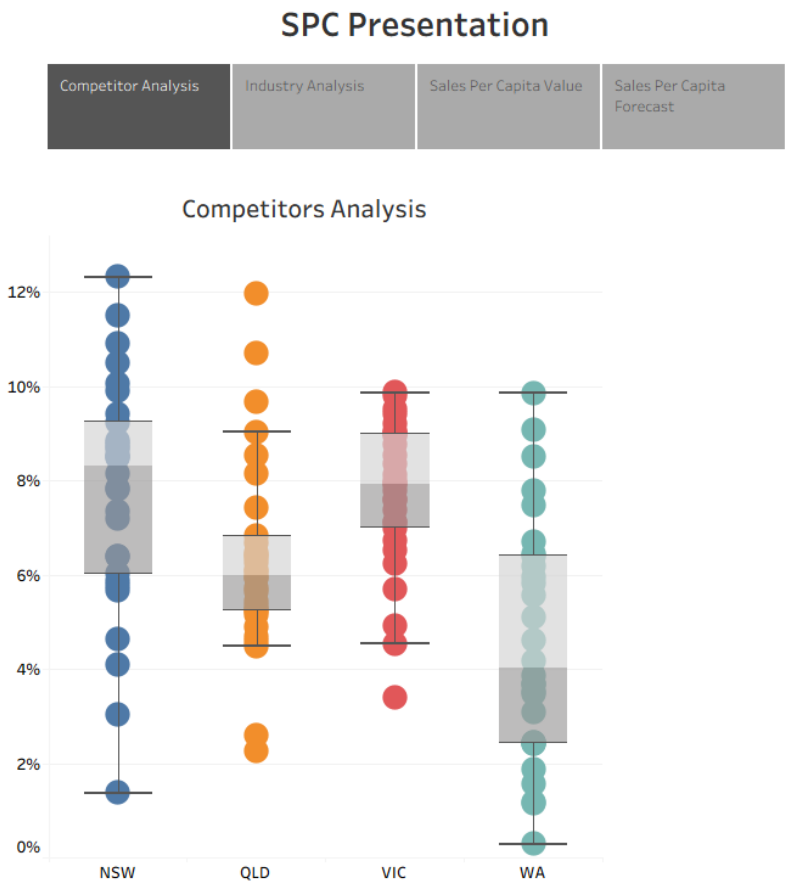
SPC Forecast -- Historical SPC $ (2000-2015) extended to 2019 with shaded confidence bands per region. NSW forecast reaches ~80-90 $, VIC ~70-80 $, QLD and WA stable.

- NSW: Upward forecast with widening confidence bands -- high growth but increasing uncertainty by 2019.
- VIC: Strong continued growth projected -- confidence bands tightest of all, suggesting most predictable trend.
- QLD: Essentially flat forecast -- consistent with the flat historical SPC trend.
- WA: Slight downward trend in forecast -- possible market saturation signal.

Tableau technique: Forecast added via Analytics pane > Forecast. Model type: Automatic (Tableau selects best exponential smoothing). Confidence interval: 95%. Forecast is shown as a separate color-shaded area appended to the historical line.

7 VIEW 5 -- SPC PRESENTATION: TABLEAU STORY WITH TAB NAVIGATION

A Tableau Story (multi-view presentation format) assembles all 4 analytical views into a single navigable dashboard with tab navigation at the top: Competitor Analysis | Industry Analysis | Sales Per Capita Value | Sales Per Capita Forecast. Each tab shows the corresponding view with consistent styling.



SPC Presentation -- Tableau Story with tab navigation. Shown: Competitor Analysis tab (box-and-whisker view). Tabs: Competitor Analysis, Industry Analysis, Sales Per Capita Value, Sales Per Capita Forecast.

8 SKILLS DEMONSTRATED

- Box-and-whisker charts: Analytics pane box plot with individual data point overlay via dual-mark technique.
- Faceted time series: Row shelf dimension creates independently-scaled panels per region.
- Trend lines: Linear trend via Analytics pane applied to multi-panel time series.
- Tableau Forecast: Built-in exponential smoothing with 95% confidence bands, extending 4 years beyond historical data.
- Tableau Stories: Multi-view narrative presentation with tab navigation -- suitable for executive dashboards.
- Per-capita normalization: Calculated field [SPC $] = SUM([Revenue]) / SUM([Population]) -- separates volume from efficiency.
- Color encoding: Consistent region color palette across all 5 views for coherent narrative flow.